University of the Philippines
Tacloban College
Second Semester AY 2026 – 2027
CMSC 122 - Data Structures and Algorithms
First Semester AY 2026-2027

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Project Requirement 1: PR 1 Project Proposal **Date**: 10/09/26

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Project title: Chase of the Cosmos - An Automated Maze Game

Project Description:

A grid-based puzzle inspired by the old Mummy Maze game, a Theseus and the Minotaur archetype, but reimagined in a cosmic environment. The player controls an astronaut navigating a 6×6 maze to reach the exit while being pursued by the big bad alien. The game implements A* pathfinding, optimizing the original BFS based logic. As a bonus challenge, the project will also optimize the original Adversary AI (Minimax) with Alpha-beta pruning, which should intuitively be more optimized within the restrictions of the game. The project will explore these optimized pathfinding tests and compare their time and space complexities against the original puzzle's known solutions.

Target user/s:

Students, myself, and players that are interested in logic puzzles. Also possibly for players that seek the nostalgia of the old Mummy Maze puzzle game, and perhaps those who wish to learn applications of pathfinding algorithms for academic purposes.

Technology Stack:

Python with libraries like Pygame for rendering and game logic. No external AI built libraries. Algorithms will be implemented manually but any inspirations or logic reused will be cited properly.